Chapter 4. The Binomial Theorem and Related Identities

No Matter How You Slice It

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This chapter pauses the enumeration of basic situations to study binomial and multinomial coefficients themselves. The proofs matter more than the identities: most show that both sides count the same objects in two different ways (**double counting**).

4.1 The binomial theorem

Motivation. Expanding $(x+y)^n$ means choosing x or y from each factor, so the coefficients count subsets. Specialising x and y produces identities, and double counting produces many more. Together they explain the patterns of the Pascal triangle.

Theorem 4.1 (Binomial theorem)

For every nonnegative integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}. \tag{4.1}$$

Proof. Multiplying out $(x+y)(x+y)\cdots(x+y)$ means choosing one summand from each of the n factors in all 2^n ways and adding the products. A choice gives $x^k y^{n-k}$ exactly when x is taken from a k-element set of factors, and there are $\binom{n}{k}$ such sets. ■

Theorem 4.2

For every positive integer n: $\sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0$.

Proof. Put $x=-1$, $y=1$ in (4.1): the left side is $0^n = 0$. ■

Theorem 4.3 (Pascal's rule)

For all nonnegative integers n, k:

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}. \tag{4.2}$$

Proof. The right side counts $(k+1)$-subsets S of $[n+1]$. If $n+1 \in S$, the rest of S is a k-subset of $[n]$ (first term); if not, S is a $(k+1)$-subset of $[n]$ (second term). ■

Theorem 4.4

For every nonnegative integer n: $2^n = \sum_{k=0}^{n}\binom{n}{k}$.

Proof. Both sides count the subsets of $[n]$: the left directly (Theorem 2.4), the right grouped by size. (Alternatively, put $x=y=1$ in (4.1).) ■

Writing $\binom{n}{i}$ as the i-th entry of row n (rows starting at 0) gives the **Pascal triangle**. Row n sums to 2^n (Theorem 4.4), each entry is the sum of the two above it (Theorem 4.3), and alternating row sums vanish (Theorem 4.2).

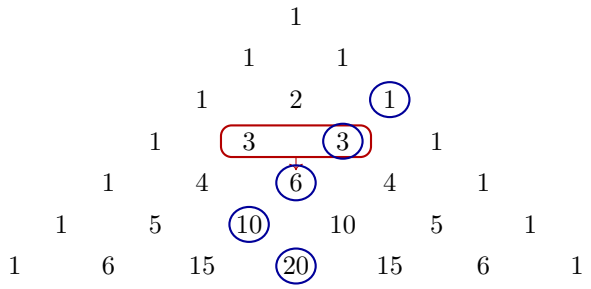


Figure. *The Pascal triangle, rows 0–6. Red: $3+3=6$ (Theorem 4.3). Blue: $1+3+6+10=20$, a "hockey stick" sum down a diagonal (Theorem 4.5).*

Theorem 4.5 (Hockey stick identity)

For all nonnegative integers $k \le n$:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}. \tag{4.3}$$

Proof. The right side counts $(k+1)$-subsets of $[n+1]$. Sort them by their largest element $k+i+1$ $(0 \le i \le n-k)$: the other k elements form a k-subset of $[k+i]$, so there are $\binom{k+i}{k}$ of them. ∎

Theorem 4.6

For every nonnegative integer n:

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}. \tag{4.4}$$

Proof.

Step 1 (double counting). Count the ways to choose a committee from n people together with its president. Choosing a k-member committee first gives $k\binom{n}{k}$, summed over k. Choosing the president first (n ways) and then any subset of the other $n-1$ people gives $n2^{n-1}$.

Step 2 (calculus, alternative). Differentiate $(x+1)^n = \sum_{k=0}^{n}\binom{n}{k}x^k$ to get $n(x+1)^{n-1} = \sum_{k\ge 1} k\binom{n}{k}x^{k-1}$, and put $x=1$. ∎

(It is not even obvious a priori that $\sum_k k\binom{n}{k}/2^{n-1}$ is an integer.)

Theorem 4.7 (Vandermonde's identity)

For all positive integers n, m, k:

$$\binom{n+m}{k} = \sum_{i=0}^{k}\binom{n}{i}\binom{m}{k-i}.$$

Proof. Classify the k-subsets of $[n+m]$ by the number i of elements taken from $[n]$: these can be chosen in $\binom{n}{i}$ ways and the other $k-i$ from $\{n+1,\dots,n+m\}$ in $\binom{m}{k-i}$ ways. ∎

Theorem 4.8

If $0 \le k \le \frac{n-1}{2}$, then

$$\binom{n}{k} \le \binom{n}{k+1}, \tag{4.5}$$

with equality iff $n = 2k+1$.

Proof. Dividing $\frac{n!}{k!(n-k)!} \le \frac{n!}{(k+1)!(n-k-1)!}$ by $n!$ and multiplying by $k!(n-k-1)!$, the inequality becomes $\frac{1}{n-k} \le \frac{1}{k+1}$, i.e. $2k+1 \le n$. The steps are reversible, and equality holds iff $n-k = k+1$. ■

Corollary 4.9

If $k \ge \frac{n-1}{2}$ (and $k+1 \le n$), then

$$\binom{n}{k} \ge \binom{n}{k+1}, \tag{4.6}$$

with equality iff $n = 2k+1$.

Proof. By Proposition 3.18(1), $\binom{n}{k} = \binom{n}{n-k}$ and $\binom{n}{k+1} = \binom{n}{n-k-1}$. With $k' := n-k-1 \le \frac{n-1}{2}$, Theorem 4.8 gives $\binom{n}{k'} \le \binom{n}{k'+1}$, which is the claim, with equality iff $n = 2k'+1 = 2(n-k-1)+1$, i.e. $n = 2k+1$. ■

A sequence $a_0, \dots, a_n$ is **unimodal** if $a_0 \le \cdots \le a_m \ge \cdots \ge a_n$ for some m. By Theorem 4.8 and Corollary 4.9, every row of the Pascal triangle is unimodal, with its peak in the middle.

4.2 The multinomial theorem

Motivation. For $(x_1 + \cdots + x_k)^n$ we choose one of k variables from each factor, so the coefficient of $x_1^{a_1} \cdots x_k^{a_k}$ counts words with prescribed letter multiplicities: the multiset permutations of Chapter 3.

Example 4.10

$$(x+y+z)^3 = x^3 + y^3 + z^3 + 3x^2y + 3x^2z + 3y^2x + 3y^2z + 3z^2x + 3z^2y + 6xyz. \tag{4.7}$$

Solution. Each of the 27 choices of one variable per factor gives a degree-3 monomial. x^2y arises in 3 ways (choose the factor giving y), x^3 in one way, and xyz in $3 \cdot 2 = 6$ ways.

Definition 4.11 (Multinomial coefficient)

For nonnegative integers with $n = \sum_{i=1}^{k} a_i$,

$$\binom{n}{a_1, a_2, \dots, a_k} := \frac{n!}{a_1!\, a_2! \cdots a_k!}. \tag{4.8}$$

For $k = 2$, $\binom{n}{a_1, a_2} = \binom{n}{a_1}$.

Theorem 4.12 (Multinomial theorem)

For all nonnegative integers n, k,

$$(x_1 + \cdots + x_k)^n = \sum_{a_1+\cdots+a_k=n} \binom{n}{a_1, \ldots, a_k} x_1^{a_1} \cdots x_k^{a_k}, \tag{4.9}$$

the sum over all k-tuples of nonnegative integers with sum n.

Proof. A term $x_1^{a_1} \cdots x_k^{a_k}$ arises from choosing x_i in exactly a_i of the n factors. Such choices correspond bijectively to words of length n with a_i letters x_i: the j-th letter is the variable chosen from the j-th factor. By Theorem 3.5 there are $\binom{n}{a_1,\ldots,a_k}$ such words. ■

Theorem 4.13

For nonnegative integers with $n = \sum_i a_i$:

$$\binom{n}{a_1, \ldots, a_k} = \binom{n}{a_1}\binom{n-a_1}{a_2} \cdots \binom{n-a_1-\cdots-a_{k-1}}{a_k} \tag{4.10}$$

(the last factor is $\binom{a_k}{a_k} = 1$).

Proof. The left side counts words with a_i copies of x_i. Build such a word by choosing the a_1 positions of x_1 ($\binom{n}{a_1}$ ways), then the a_2 positions of x_2 among the remaining $n - a_1$, and so on. ■

4.3 When the exponent is not a positive integer

Motivation. Generating functions (Chapter 8) need expansions like $(1-4x)^{-1/2}$. Extending $\binom{m}{k}$ to real m turns the binomial theorem into a power series, Newton's binomial series.

Definition 4.14 (General binomial coefficients)

For real m and integer $k \geq 0$: $\binom{m}{0} := 1$ and $\binom{m}{k} := \frac{m(m-1)\cdots(m-k+1)}{k!}$ for $k > 0$.

Theorem 4.15 (Binomial theorem, general version)

For every real m and $|x| < 1$,

$$(1+x)^m = \sum_{n \geq 0} \binom{m}{n} x^n.$$

Proof [Supplement]. The book derives this from Taylor's theorem, since the n-th derivative of $(1+x)^m$ at 0 is $m(m-1)\cdots(m-n+1)$. Here is a complete argument.

Step 1 (convergence). If $m \in \mathbb{N}$ the series is a finite sum and (4.1) applies. Otherwise all $\binom{m}{n} \neq 0$ and $\left|\binom{m}{n+1}x^{n+1} / \binom{m}{n}x^n\right| = \frac{|m-n|}{n+1}|x| \to |x| < 1$. By the ratio test the series converges on $(-1, 1)$ and defines a differentiable f with $f(0) = 1$, differentiable term by term.

Step 2 (differential equation). Using $(n+1)\binom{m}{n+1} = (m-n)\binom{m}{n}$,

$$(1+x)f'(x) = \sum_n \left[(n+1)\binom{m}{n+1} + n\binom{m}{n}\right]x^n = \sum_n m\binom{m}{n}x^n = mf(x).$$

Step 3. Then $\frac{d}{dx}\left[f(x)(1+x)^{-m}\right] = (1+x)^{-m-1}\left[(1+x)f' - mf\right] = 0$, so $f(x)(1+x)^{-m}$ is constant, equal to its value 1 at $x = 0$. ■

If m is a positive integer, $\binom{m}{n} = 0$ for $n > m$ and we recover (4.1).

Example 4.16

$$\frac{1}{\sqrt{1-4x}} = \sum_{n\geq 0} \binom{2n}{n} x^n.$$

Solution. By Theorem 4.15, $(1-4x)^{-1/2} = \sum_n \binom{-1/2}{n}(-4x)^n$ and

$$\binom{-1/2}{n} = \frac{(-\frac{1}{2})(-\frac{3}{2})\cdots(-\frac{2n-1}{2})}{n!} = (-1)^n \frac{(2n-1)!!}{2^n n!}, \tag{4.11}$$

where $(2n-1)!! = 1\cdot 3\cdots(2n-1)$ is the **semifactorial** and $(-1)!! := 1$. The coefficient of x^n is therefore $\frac{2^n(2n-1)!!}{n!}$. Multiplying numerator and denominator by $n!$ and using $2^n n! = 2\cdot 4\cdots 2n$, this equals $\frac{(2n)!}{n!\,n!} = \binom{2n}{n}$, which also holds for $n = 0$.

4.4 Selected exercises

Motivation. The book exercises below prove identities and inequalities for binomial coefficients in three typical ways: double counting, comparing consecutive ratios, and a bijection on lattice paths.

Exercise 4.4 (Choosing two disjoint subsets)

Let k, m, n be nonnegative integers with $k + m \leq n$. Prove that

$$\binom{n}{m}\binom{n-m}{k} = \binom{n}{k}\binom{n-k}{m}.$$

Solution. Count the ordered pairs (M, K) of disjoint subsets of $[n]$ with $|M| = m$ and $|K| = k$.

Step 1 (M first). Choose M in $\binom{n}{m}$ ways, then K among the $n-m$ remaining elements in $\binom{n-m}{k}$ ways. Every pair arises exactly once, so there are $\binom{n}{m}\binom{n-m}{k}$ pairs.

Step 2 (K first). Choosing K first and then M in $[n] \setminus K$ gives $\binom{n}{k}\binom{n-k}{m}$ for the same number.

Comparing the two counts proves the identity. (Both sides equal $\binom{n}{m,k,n-m-k}$, by Theorem 4.13.) ■

Exercise 4.21 (Log-concavity of binomial coefficients)

(a) Prove that for positive integers $k \leq n-1$,

$$\binom{n}{k-1}\binom{n}{k+1} \leq \binom{n}{k}^2.$$

(b) (Exercise 4.23.) A sequence $a_0, \ldots, a_n$ of positive reals is *log-concave* if $a_{i-1}a_{i+1} \leq a_i^2$ for $1 \leq i \leq n-1$. Prove that a log-concave sequence is unimodal.

Solution.

(a) From $\binom{n}{j} = \frac{n!}{j!(n-j)!}$ we get, for $1 \le k \le n-1$,

$$\frac{\binom{n}{k-1}}{\binom{n}{k}} = \frac{k}{n-k+1}, \qquad \frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}.$$

Multiplying, $\binom{n}{k-1}\binom{n}{k+1}/\binom{n}{k}^2 = \frac{k(n-k)}{(k+1)(n-k+1)} < 1$, since $k < k+1$ and $0 < n-k < n-k+1$. This is the claim, even with strict inequality.

(b)

Step 1 (ratios). Let $r_i = a_i/a_{i-1}$ for $1 \le i \le n$. Dividing $a_{i-1}a_{i+1} \le a_i^2$ by the positive number $a_{i-1}a_i$ gives $r_{i+1} \le r_i$, so $r_1 \ge r_2 \ge \cdots \ge r_n$.

Step 2 (the peak). Let m be the largest index $i \in [n]$ with $r_i \ge 1$, and $m = 0$ if there is none. For $i \le m$ we have $r_i \ge r_m \ge 1$, i.e. $a_{i-1} \le a_i$; for $i > m$ we have $r_i < 1$, i.e. $a_{i-1} > a_i$. Hence $a_0 \le \cdots \le a_m > a_{m+1} > \cdots > a_n$, and the sequence is unimodal.

Together, (a) and (b) give a second proof that the rows of the Pascal triangle are unimodal. ■

Exercise 4.24 (Catalan numbers by reflection)

Let C_n be the number of north-eastern lattice paths (steps $(1,0)$ and $(0,1)$) from $(0,0)$ to (n,n) that never go above the diagonal $x = y$. Prove that

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}.$$

Solution. A path from $(0,0)$ to (n,n) has n east and n north steps, so there are $\binom{2n}{n}$ such paths (choose the positions of the east steps). Along a path let $d = y - x$; each step changes d by ± 1. A path is *bad* if it goes above the diagonal, i.e. if it visits a point with $d = 1$. We show that the bad paths are equinumerous with the paths from $(-1,1)$ to (n,n), of which there are $\binom{2n}{n-1}$ ($n+1$ east and $n-1$ north steps).

Step 1 (reflection). Let $\varphi(x,y) = (y-1, x+1)$ be the reflection in the line $y = x+1$. It fixes the points with $d = 1$, maps a point with d to one with $2-d$, and turns an east step into a north step and vice versa. Also $\varphi(0,0) = (-1,1)$.

Step 2 (bad paths to paths from $(-1,1)$). For a bad path P let t be its first point with $d = 1$. Reflect the part of P from $(0,0)$ to t by φ and keep the rest. The result is a lattice path from $(-1,1)$ to (n,n).

Step 3 (the inverse). A lattice path Q from $(-1,1)$ to (n,n) starts with $d = 2$ and ends with $d = 0$, and d moves by ± 1, so Q visits a point with $d = 1$. Reflect Q up to its first such point t. Since φ maps points with $d \ne 1$ to points with $d \ne 1$, the first point with $d = 1$ is the same before and after reflecting. Hence the two operations are inverse to each other, and Step 2 is a bijection.

Step 4 (conclusion). $C_n = \binom{2n}{n} - \binom{2n}{n-1}$. Since $\binom{2n}{n-1} = \binom{2n}{n} \cdot \frac{n}{n+1}$ (by the ratio in Exercise 4.21(a)), $C_n = \binom{2n}{n}\left(1 - \frac{n}{n+1}\right) = \frac{1}{n+1}\binom{2n}{n}$. For $n = 0$ both sides equal 1.

This is the formula found with generating functions in Chapter 8. ■